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1 February 1989
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MILITARY SPECIFICATION

PLASTIC SHEET, ACRYLIC, MODIFIED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers optical quality transparent, cast acrylic sheet material which is superior to conventional acrylic material in craze and heat resistance.

*1.2 Classification. The acrylic sheet shall be of the types, classes, grades and thicknesses as specified (see 6.2.1).

*1.2.1 Types (See 3.1.1).

Type I - For use "as cast" and in stretching operations.
Type II - For use "as cast" only. Not suitable for stretching.

*1.2.2 Classes.

Class 1 - Standard moisture absorption resistance.
Class 2 - Improved moisture absorption resistance.

1.2.3 Sizes. Sizes shall be as specified in 6.7.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 53), Naval Air Engineering Center, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.1 Specifications, standards and handbooks. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

NN-P-530	-	Plywood, Flat Panel.
TT-I-735	-	Isopropyl Alcohol.
TT-T-548	-	Toluene, Technical.
TT-W-572	-	Wood-Preservative, Water-Repellent.
PPP-B-585	-	Box, Wood, Wirebound.
PPP-B-591	-	Boxes, Shipping, Fiberboard, Wood-Cleated.
PPP-B-601	-	Boxes, Wood, Cleated Plywood.
PPP-B-621	-	Box, Wood, Nailed and Lock Corner.

MILITARY

MIL-L-10547	-	Liner, Case and Sheet, Overwrap, Water Vaporproof or Waterproof, Flexible.
MIL-P-25690		Plastic, Sheets and Parts, Modified Acrylic Base, Monolithic, Crack Propagation Resistant

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 542	-	Index of Refraction of Transparent Organic Plastics.
ASTM D 570	-	Water Absorption of Plastics.
ASTM D 635	-	Rate of Burning and/or Extent of Time of Burning of Self-Supporting Plastics in a Horizontal Position.
ASTM D 637	-	Surface Irregularities of Flat Transparent Plastic Sheets.
ASTM D 638	-	Tensile Properties of Plastics.
ASTM D 648	-	Deflection Temperature of Plastics Under Flexural Load.
ASTM D 696	-	Coefficient of Linear Thermal Expansion of Plastics.
ASTM D 792	-	Specific Gravity and Density of Plastics by Displacement.
ASTM D 1003	-	Haze and Luminous Transmittance of Transparent Plastics.
ASTM D 1718	-	Isobutyl Acetate.
ASTM E 831	-	Linear Thermal Expansion of Solid Materials by Thermodilatometry.
ASTM G 26	-	Operating Light-Exposure Apparatus (Xenon-Arc Type) with and without Water for Exposure of Non-metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Nongovernment standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein , the text of this document takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The acrylic sheets furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for award of contract (see 4.3 and 6.3).

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*3.1.1 Types. The manufacturer shall determine the suitability of these materials for producing stretched material conforming to MIL-P-25690 prior to submission of samples for qualification and shall classify the material either type I or Type II, accordingly. Type I sheets can be used in either "as cast" applications or in stretching. Type II sheets shall be used "as cast" only.

3.2 Materials. The plastic sheet shall be an acrylic type. The manufacturer is given a wide range in the selection of raw acrylic type material and in the process of manufacture, provided the sheet material furnished is a transparent plastic conforming to all requirements of this specification as applicable, and is suitable for the intended use.

3.3 Dimensions.

3.3.1 Length and width. Unless otherwise specified, dimensions of the sheets shall be as specified by applicable drawings or specifications. Unless otherwise specified, a tolerance of ± 0.063 inch (1.6mm) shall be permitted on length and width dimensions for trimmed sheets at 23° ± 1°C (73.5° ± 2° F).

3.3.2 Thickness. Thickness shall be as specified in Table I. The actual thickness, when examined as specified in 4.6.14, shall be within the tolerances in Table I. Tolerances for those thicknesses not in Table I shall not exceed the tolerance for the next larger thickness.

3.4 Color. Unless otherwise specified by the acquisition activity (see 6.2.1), the material shall contain no added dyes or pigments.

3.5 Specific gravity. The specific gravity of the conditioned material shall be 1.19 ± 0.01, when determined as specified in 4.6.1.

*3.6 Water absorption. When determined in accordance with 4.6.1, the water absorption of the conditioned material from each class shall not exceed the values given in Table I.

*3.7 Long term water absorption. When tested as specified in 4.6.15, Class 2 acrylic materials shall not absorb more than 2.90 percent water.

3.8 Flammability. When tested as specified in 4.6.1, the rate of burning for a 0.500 inch (1.25 cm) width of material shall not exceed the values given in Table I.

3.9 Thermal expansion. The coefficient of thermal expansion shall not exceed 0.00010 inch/inch/degree Centigrade (0.000055 inch/inch/degree Fahrenheit), when tested in accordance with 4.6.2.

3.10 Formability. Formability shall be determined as specified in 4.6.3. The material shall be suitable for forming into hemispheric shapes with an outside diameter of 10 ± 0.1 inches (25 ± 0.2 cm) and a draw of at least 4.5 inches (11.5 cm).

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TABLE I. Requirements which vary with thickness. 1/

Nominal thickness inch	Thickness tolerances inch			Heat deflection temperature				Water absorption percent (max)	Flammability inch per minute (max)	Warpage after accelerated weathering inch (max)
				Max		Min				
	Sheet up to 36 inch by 60 inch	sheet larger than 36 by 60 inch up to 60 by 80 inch	sheet larger than 60 by 80 inch up to 72 by 100 inch	°C	°F	°C	°F			
0.060	±0.012	±0.020	--	112	236	87	189	1.00	1.60	0.030
0.080	±0.012	±0.020	--	114	240	89	192	0.80	1.55	0.020
0.100	±0.012	±0.020	--	116	244	91	196	0.70	1.45	0.020
0.125	±0.015	±0.020	±0.030	118	248	93	199	0.65	1.40	0.015
0.150	±0.017	±0.020	±0.030	119	250	94	201	0.60	1.37	0.015
0.187	±0.020	±0.023	±0.030	121	254	96	205	0.50	1.35	0.015
0.220	±0.023	±0.025	±0.030	122	252	97	207	0.45	1.35	0.015
0.250	±0.025	±0.030	±0.035	123	253	98	209	0.40	1.35	0.015
0.312	±0.030	±0.035	±0.040	124	255	99	211	0.36	1.35	0.015
0.375	±0.035	±0.040	±0.045	125	257	100	212	0.30	1.30	0.015
0.417	±0.040	±0.045	±0.045	125	257	100	212	0.28	1.30	0.015
0.500	±0.040	±0.045	±0.050	126	259	101	214	0.25	1.30	0.015
0.625	±0.050	±0.050	±0.055	126	259	101	214	0.23	1.20	0.015
0.750	±0.050	±0.050	±0.060	126	259	101	214	0.20	1.10	0.015
0.875	±0.050	±0.050	±0.070	126	259	101	214	0.20	1.10	0.015
1.000	±0.050	±0.050	±0.075	127	261	102	216	0.20	1.00	0.015
1.250	±0.063	±0.063	±0.095	127	261	102	216	0.20	1.00	0.015
1.500	±0.075	±0.075	±0.112	127	261	102	216	0.20	1.00	0.015
1.750	±0.088	±0.088	±0.131	127	261	102	216	0.20	1.00	0.015
2.000	±0.100	±0.100	±0.150	127	261	102	216	0.20	1.00	0.015
2.250	±0.113	±0.113	±0.168	127	261	102	216	0.20	1.00	0.015
2.500	±0.117	±0.117	±0.175	127	261	102	216	0.20	1.00	0.010
2.000	±0.120	±0.120	±0.190	127	261	102	216	0.20	1.00	0.010
3.500	±0.140	±0.140	±0.210	127	261	102	216	0.20	1.00	0.010
4.000	±0.160	±0.160	±0.225	127	261	102	216	0.20	1.00	0.010

1/ Conversion inch to mm = inch X 25.4.

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3.11 Internal strain. The dimensional change after testing as specified in 4.6.4 shall not exceed 1 percent. Large values of dimensional change indicate the relief of high internal stress.

3.12 Flexural deformation temperature. The flexural deformation temperature of the acrylic sheet shall be within the range specified for the thickness in Table I when tested as specified in 4.6.5.

3.13 Mechanical properties.

3.13.1 Tensile strength. The tensile strength of materials as received shall be not less than 9,000 psi (62.1 MPa), when tested in accordance with 4.6.6.1 (see 4.6.6 for specimen preparation).

3.13.2 Elongation. The mean elongation immediately before fracture shall be not less than 2 percent as determined in 4.6.6.2.

3.13.3 Warpage after accelerated weathering. The warpage of the plastic after accelerated weathering shall not exceed the value given in Table I as determined in 4.6.13.1. The specimen shall show no evidence of cracking, crazing or surface instability.

3.14 Instruction sheet. An instruction sheet shall be furnished in each shipping container (see 5.2). Specific information or reference shall be made to the limitations of the material and the necessary precautions to be observed in handling, storing, cutting, drilling, machining, forming, bending, cementing, abrading, polishing and cleaning. Instruction sheets shall be approved by the qualifying activity (see 4.3.3).

3.15 Optical properties.

3.15.1 Property. Materials as received and after exposure to accelerated weathering shall conform to the requirements of Table II.

3.15.2 Resistance to weathering. After exposure to accelerated weathering at the conditions specified in 4.6.13, specimens shall show no evidence of cracking, crazing or other indications of surface instability that could affect visibility, except that scratches incident to exposure may be disregarded.

3.15.3 Optical uniformity.

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TABLE II. Special physical properties.

Characteristics	Conditioning of specimen		Test paragraph
	as received	After accelerated weathering	
Ultraviolet transmittance, 290 to 330 mu, percent, maximum:	2.0	-	4.6.7
Index of refraction	1.49 ± 0.01	-	4.6.8
Luminous transmittance of colorless sheet, percent, minimum:			4.6.9
Thickness, inch (mm 1/):			
0.060 through 0.187	91	90	
Over 0.187 through 0.312	90	89	
Over 0.312 through 0.417	89	87	
Over 0.417 through 1.250	88	86	
Over 1.250 through 2.250	88	85	
Over 2.250 through 3.000	85	82	
Over 3.000 through 4.000	80	80	
Colorless sheet, haze, percent, max.	3.0	4.0	4.6.9

1/ Conversion to mm = inch X 25.4.

3.15.3.1 Minor optical defects. The total number of minor optical defects in material which is 0.500 inch (12.5 mm) or less in thickness shall not exceed a limit determined by dividing the area of the sheet, measured in square feet (square meters), by 4. The total number of minor optical defects in material over 0.500 inch (12.5 mm) in thickness shall not exceed 1 per square foot (square meter). Minor defects include any embedded particles, bubbles or scratches which reduce visibility through the plastic, and those localized imperfections which cause a variation in angular deviation of more than 5 minutes within a distance of not more than 20 inches (0.51 m) on the screen when tested as specified in Table III. It is not intended that the entire sheet be quantitatively surveyed for such variation in deviation but that localized imperfections which are suspected of being detrimental be evaluated by means of this test. Blemishes which do not individually reduce visibility through the plastic shall be disregarded unless they are grouped in an objectionable pattern. Minor defects within 1 inch (25.4 mm) from the nominal trimmed sheet edge shall be disregarded.

3.15.3.2 Angular deviation. The material shall contain no major defects. Major defects are defined as any variations in the material which cause angular deviations on either side of the undeviated position in excess of the limits specified in Table III.

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TABLE III. Angular deviation requirements.

Sheet thickness 1/	Limits of permissible deviation 2/
0.060 inch through 0.220 inch	7 minutes at any location more than 1 inch from the edge of the sheet.
Over 0.220 inch through 0.250 inch	7 minutes at any location more than 3 inches from the edge of the sheet and 9 minutes between 3 inches and 1 inch of the edge.
Over 0.250 inch through 0.375 inch	7 minutes at any location more than 3 inches from the edge and 12 minutes between 3 inches and 1 inch of the edge.
Over 0.375 inch through 0.500 inch	7 minutes at any location more than 3 inches from the edge and 14 minutes between 3 inches and 1 inch of the edge.
Over 0.500 inch through 1.000 inch	12 minutes at any location more than 3 inches from the edge and 20 minutes between 3 inches and 1 inch of the edge.
Over 1.000 inch through 2.500 inch	20 minutes at any location more than 3 inches from the edge and 25 minutes between 3 inches and 1 inch of the edge.
Over 2.500 inch through 4.000 inch	30 minutes at any location more than 3 inches from edge and 35 minutes between 3 inches and 1 inch of the edge.

1/ Conversion to mm = inch X 25.4.
2/ Major defects within 1 inch from the edge shall be disregarded.

3.15.4 Thermal stability. The material shall show no evidence of blistering, crazing or other evidence of thermal instability when tested as specified in 4.6.10.

3.15.5 Properties of colored material. When colored material is furnished, the limits of spectral transmittance and haze before and after accelerated weathering shall be as specified by the acquiring activity (see 6.2.1).

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* 3.15.6 Stress to craze. Stress to craze values when tested in accordance with 4.4.2, shall be as specified below:

	Isopropanol (TT-I-735)		Toluene/Isobutylacetate (TT-T-548/ASTM D 1718)	
	Dry	Wet	Dry	Wet
Class 1, psi	2500	1500	2000	1500
Class 2, psi	2500	2000	2000	1600

* 3.16 Identification of product. The protective covering (see 5.1) of each individual sheet shall be marked with the specification number, Type, Class, nominal thickness, and the manufacturer's code for his approved product. The protective covering shall be distinctly marked at intervals of 1 foot (0.3 m) (see 5.3.1).

3.17 Workmanship. Workmanship shall be in accordance with high-grade practice for this type of product. The finished sheet shall be free from such bubbles, striae and other defects as would render the material unfit for the purpose of viewing objects through it. Since this specification does not apply to formed, molded or fabricated parts, workmanship beyond the production of polished plane sheet materials is not included.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).

b. Qualification retention (see 4.3.4).

c. Quality conformance inspection (see 4.4).

4.3 Qualification. Qualification inspection shall consist of all the tests specified in Table IV.

TABLE IV. Qualification tests and inspections.

Characteristics	Paragraph	
	Requirement	Test
Specific gravity	3.5	4.6.1
Water absorption	3.6	4.6.1
Long term water absorption	3.7	4.6.15
Flammability	3.8	4.6.1
Thermal expansion	3.9	4.6.2
Formability	3.10	4.6.3
Internal strain	3.11	4.6.4
Flexural deformation	3.12	4.6.5
Tensile strength	3.13.1	4.6.6
Elongation	3.13.2	4.6.6
Ultraviolet transmittance	3.15.1	4.6.7
Index of refraction	3.15.1	4.6.8
Luminous transmittance and haze	3.15.1	4.6.9
Thermal stability	3.15.4	4.6.10
Optical uniformity	3.15.3	4.6.11
Crazing	3.15.6	4.6.12
Resistance to weathering	3.15.2	4.6.13
Warpage	3.13.3	4.16.13.1
Workmanship	3.16	All

4.3.1 Qualification test sample.

* 4.3.1.1 Sheet sample. To obtain qualification approval for all thicknesses, the sample shall consist of at least 10 square feet (1 square meter) of trimmed sheets in each of 0.060 inch, 0.250 inch, 0.500 inch, 1.00 inch, 2.25 inch and 4.00 inch thicknesses. Qualification of two consecutive ranges, i.e. 0.060 and 0.250 inch, automatically extends qualification to all sizes within the two ranges. Individual sheet dimensions shall be at least 12 by 18 inches. If a manufacturer requests approval for a specific thickness, he shall submit 10 square feet of that thickness. A portion of the sheet sample shall be used by the manufacturer to prepare the specimens required in 4.3.1.2, with the remaining sheet material submitted to the qualifying laboratory (see 4.3.2).

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* 4.3.1.2 Test specimen samples. A portion of the sheet sample (see 4.3.1.1) shall be used by the manufacturer to prepare sufficient test specimens to complete the tests specified for flammability (4.6.1), tensile properties (see 4.6.6), flexural deformation (see 4.6.5), thermal expansion (see 4.6.2), crazing (see 4.6.12), formability (see 4.6.3) and ultraviolet transmittance (see 4.6.7). Preparation of these specimens requires the sample to be machined and polished to specific thicknesses required by the applicable test.

* 4.3.2 Forwarding of qualification samples. The qualification samples from 4.3.1.1 and 4.3.1.2 shall be forwarded to the Commander, Naval Air Development Center, Warminster, PA 18974, Attention: Aircraft and Crew Systems Technical Directorate (Code 6061). The samples shall be segregated by thickness (machined samples shall be identified as to original thickness). The samples shall be plainly and durably marked with the following information:

Sample for qualification tests

PLASTIC SHEET, ACRYLIC, MODIFIED
Type and Class
Plant address (where manufactured)
Submitted by (name) (date) for qualification testing in accordance with the requirements of MIL-P-8184E under (reference authorizing letter) (see 6.3).

4.3.3 Manufacturer's data. Two copies of the manufacturer's test report shall be submitted with the samples of 4.3.1. The report shall contain numerical test data, where applicable, showing that the material submitted for the qualification inspection conforms to the requirements of this specification. In addition, two copies of the manufacturer's instruction sheet (see 3.13) shall be furnished at this time.

4.3.4 Retention of qualification. In order to retain qualification of a product approved for listing on the Qualified Products List (QPL), the manufacturer shall verify by certification to the qualifying activity that the manufacturer's product complies with the requirements of this specification. Qualifying activity shall initiate verification by certification in two year intervals. The Government reserves the right to re-examine the qualified product whenever deemed necessary to determine that the product continues to meet any or all of the specification requirements.

4.4 Quality conformance inspection.

* 4.4.1 Lot formation. Unless otherwise specified, a lot shall consist of the total number of sheets of one type, class and thickness forming part of the contract or order, submitted for delivery at one time.

4.4.2 Sampling.

4.4.2.1 Visual inspection sampling. The samples for inspections of 4.4.3.1 and 4.4.3.2 shall be a quantity of plastic sheets randomly selected from each lot in accordance with the procedures of MIL-STD-105 and Table V. The sample unit shall be one plastic sheet.

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4.4.2.2 Packaging inspection sampling. A quantity of shipping containers fully prepared for shipment, just prior to closure, shall be selected in accordance with MIL-STD-105 and Table V. The lot size, for purposes of this inspection, shall be the total number of shipping containers. The sample unit shall be one shipping container. Inspection shall be in accordance with 4.4.3.3.

4.4.2.3 Physical and mechanical testing. The sample shall be randomly selected from each lot in accordance with MIL-STD-105 and Table V, except that no fewer than 3 sample units shall be taken. The sample unit shall be one sheet of at least 4 square feet with a minimum dimension of 12 inches. Each sample unit selected shall be inspected as specified in 4.4.3.4.

TABLE V. Inspection levels and acceptable quality levels (AQL).

Paragraph		Inspection level	AQL 1/
Sampling	Inspection		
4.4.2.1	4.4.3.1	II	1.0 major, 4.0 minor
4.4.2.1	4.4.3.2 2/	II	1.0
4.4.2.2	4.4.3.3	S-2	4.0
4.4.2.3.1	4.4.3.4	S-3	4.0

1/ Percent defective.

2/ Sample units selected for 4.4.3.1 may be used in 4.4.3.2.

4.4.3 Inspections.

4.4.3.1 Appearance and workmanship. Each sample unit selected in accordance with 4.4.2.1 shall be inspected to the characteristics in Table VI. Minor optical imperfections and blemishes shall be evaluated as specified in 3.14.3.1 before being scored as defects.

4.4.3.2 Dimensional. Each sample unit selected in 4.4.2.1 shall be inspected for dimensional defects in accordance with Table VII.

4.4.3.3 Packaging inspection. Samples selected as specified in 4.4.2.2 shall be examined for conformance to Table VIII and Section 5 of this specification. In addition, shipping containers fully prepared for delivery shall be examined for closure defects.

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TABLE VI. Quality conformance appearance inspection.

Examine	Defect	Classification of defects
Appearance	Any visible haze.	Minor
	Not free of dyes or pigments.	Minor
	Scratches, cut, cracked or otherwise damaged.	Major
	Protective sheets mark surfaces when removed, not easily peeled.	Minor
	Imbedded dirt, foreign material.	Major
Workmanship	Rough, uneven edges.	Minor
	Optical defects such as striae, bubbles, blisters, "fish-eyes" and other blemishes, except as described as minor defects.	See 3.14.3.1

TABLE VII. Quality conformance dimensional inspection.

Examine	Defect
Length and width	Not as specified by the applicable drawings or specifications for stock size sheets. (Unless otherwise specified, a tolerance of ± 1/4 inch shall apply). Trimmed sheets varies by more than ± 0.06 inch from dimension specified.
Thickness	Varies by more than tolerances indicated in Table I for applicable size sheet.

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TABLE VIII. Quality conformance packaging inspection.

Examine	Defect
Packaging	Individual sheets not packaged as specified.
Packing	Packing material not as specified. Not in accordance with contract requirements.
	Container not as specified, closure not accomplished by specified or required methods or materials.
	Inadequate application of components, such as incomplete closures of case liners, container flaps, loose or inadequate strappings, bulged or distorted containers.
Instruction sheet	Missing or not as specified (see 3.13).
Count	Less than specified or indicated quantity.
Weight	Gross or net weight exceeds specified requirements.
Markings	Interior or exterior markings (as applicable) omitted, illegible, incorrect, incomplete, or not in accordance with contract requirements (see 5.3).

4.4.3.4 Physical and mechanical inspection. Samples selected as specified in 4.4.2.3 shall be tested to the requirements of Table IX The number of specimens for each sample unit and reporting of results shall be as specified therein. Failure of any requirement specified in Table IX shall be cause to reject the production lot represented by the sample.

4.5 Test conditions.

4.5.1 Standard conditions. Standard conditions shall be 25° ± 1° C (77° ± 2° F) and a relative humidity of 50 ± 5 percent. Unless otherwise specified, all tests and examinations shall be conducted at standard conditions.

4.5.2 Reporting of test results. Unless otherwise specified in the applicable test method, all results shall be reported as the average of the number of specimens being tested. Each individual value shall also be reported.

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TABLE IX. Quality conformance - physical and mechanical inspection.

Characteristic 1/	Paragraph		No. of determinations per sample unit	Results reported as:	
	Requirement	Test		Pass or fail	Average numerically to the nearest 2/
Stress to craze toluene/isobutyl acetate wet only	3.14.6	4.6.12	2	X	
Internal strain	3.10	4.6.4	2	X	
Flexural deformation temperature	3.12	4.6.6	2	-	0.1 degree C
Angular deviation	3.15.3.2	4.6.12.2	2	-	1 minute

1/ Angular deviation shall be completed prior to internal strain and flexural deformation.

2/ Averages and test results shall be reported as specified in 4.3.2.

4.6 Test methods.

4.6.1 ASTM methods. Those requirements tested entirely to ASTM methods are identified in Table X. All other ASTM methods herein are included as part of a test method.

* TABLE X. ASTM methods.

Property	Requirement	ASTM method
Specific gravity	3.5	D 792
Water absorption	3.6	D 570 1/
Flammability	3.7	D 635

1/ Condition 1 hour at 105° to 110°C (221° to 230°F), then 24 hours immersion in water at room temperature.

4.6.2 Thermal expansion. Two specimens, 0.250 inch thick, shall be tested in accordance with ASTM D 696 or ASTM E 831 to determine conformance with 3.9.

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4.6.3 Formability. Two specimens shall be formed to determine conformance to 3.9. Forming conditions shall be in accordance with the manufacturer's instructions. Material with a thickness up to 0.500 inch shall be tested in the as-cast form. Material of greater thickness shall be machined to 0.500 inch prior to testing.

4.6.4 Internal strain. Two conditioned 12-by-18 inch sheets shall be tested. Each sheet shall be considered as a 12-by-12 inch specimen supported by the remainder of the sheet. Two fine lines shall be scribed at right angles crossing the center of the 12-by-12 inch area. Finely scribed gage marks shall then be placed 2 inches from the edge of the 12-by-12 inch area on each of these lines. The distance between each pair of gage marks shall be measured to the nearest 0.06 inch (0.15 cm) and the data recorded. Each sheet shall be hung by one short length in a circulating air oven at 160° ± 10°C (320°± 18°F) for the time indicated below:

Nominal sheet thickness, inches	Minimum heating time, minutes
0.250 and less	16
0.375	25
0.500	33
0.750	55
1.000	79
1.500	136
2.000	203
2.500	240
3.000	300
3.500	350
4.000	400

After removal from the oven, the specimens shall be permitted to cool to standard testing conditions (see 4.5.1) while hanging vertically. The distance between each pair of gage marks shall be remeasured. The dimensional change shall then be computed as the percent change in distance between the gage marks based on the first measurement. The average of the four values shall be reported.

4.6.5 Flexural deformation temperature. Two specimens shall be tested in accordance with ASTM D 648, except as follows: The thickness of the sample being tested shall become the width of the specimen. Those thicknesses not in the range specified in ASTM D 648 shall be plied or machined. The unmachined surface shall be on a side. The load shall be calculated to give a maximum fiber stress of 264 psi. Each value shall be reported.

4.6.6 Tensile properties. Tensile and elongation specimens shall be prepared from as received thicknesses up to and including 0.500 inch (1.25 cm). All other thicknesses shall be machined to 0.500 inch (1.25 cm) using the as-cast surface as the face of the specimen. The machined surface shall be polished. The specimens shall be Type II specimens as described in ASTM D 638. Prior to testing, machined specimens shall be annealed for at least 2 hours at 90° C (194° F) and slowly cooled (less than 15° C (27° F) per hour) to 23° C (74° F) to relieve stresses induced during machining.

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4.6.6.1 Tensile strength. Five specimens shall be tested in accordance with ASTM D 638.

4.6.6.2 Elongation. Elongation shall be determined just before break. Procedures shall be in accordance with ASTM D 638.

4.6.7 Ultraviolet transmittance. The spectral transmittance shall be determined using a monochromator having a band width of 10 nanometers or less, and a photometer having a reproducibility of ± 1 percent using a 0.250 inch (0.63 cm) thick specimen.

4.6.8 Index of refraction. Three specimens shall be tested in accordance with the refractometer procedure of ASTM D 542. Requirements are for clear, colorless material.

4.6.9 Luminous transmittance and haze. Three specimens, 2 by 3 inches (5 by 7 cm), shall be examined as specified in ASTM D 1003, procedure A or B, for light transmission and haze. The same specimens shall be subjected to accelerated weathering (see 4.6.13). After weathering, immerse the specimens in distilled water for not longer than ten seconds, remove surface moisture by blotting, then re-examine for transmittance and haze.

4.6.10 Thermal stability. Two conditioned 12 by 18 inch (30 by 45 cm) sheets shall be tested. Each sheet shall be hung in a circulating air oven, at 180°± 5° C (356° ± 9° F) for 2 hours + 5,-0 minutes. After removal from the oven, the specimens shall be permitted to cool to standard conditions (see 4.5.1) while hanging vertically, then visually examined for conformance to 3.14.4.

4.6.11 Optical uniformity.

4.6.11.1 Localized imperfections. Local areas which, upon visual examination, are suspected of containing localized optical imperfections exceeding the limits of 3.15.3.1 shall be tested in accordance with ASTM D 637.

*4.6.11.2 Angular deviation. The angular deviation shall be determined in accordance with ASTM D 637 except that the displacement factor (angular deviation minutes) shall be determined by multiplying the maximum image movement in inches on the screen by 12. Each sheet shall be examined, then rotated 90 degrees, and re-examined for conformance to 3.14.3.2.

*4.6.12 Crazing.

4.6.12.1 Conditioning for specimens. Twenty specimens 1 by 15 by thickness up to and including 0.500 inch (2.5 by 38.1 by 1.25 cm) shall be annealed for 2 hours at 120° ± 1°C (248° ± 2° F). Thicknesses greater than 0.500 (1.25 cm shall be machined to 0.5 inch (1.25 cm). Immediately following the annealing procedure, ten specimens shall be conditioned 48 hours at standard testing conditions (see 4.5.1). The remaining ten specimens shall be cooled in air for 1 hour, immersed in water at 49° ± 1°C (120° ± 2° F) for 24 hours, followed by immersion in water at 23° ± 1°C (73° ± 2° F) for 2 to 3 hours. Water immersed specimens shall be tested within 15 minutes of removal from water. Other specimens shall be tested as soon as possible after completion of the conditioning period.

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4.6.12.2 Procedure. Each specimen shall be measured, taped with a 0.125 inch (0.32 cm) wide black matte tape (Chartpak) or equivalent, (see Figure 1), then set up as a cantilever beam under the load derived from the equation in Figure 1. Place the specimens under the load for 10 minutes before adding the fluids. Ten specimens shall be tested with isopropyl alcohol (TT-I-735) and ten specimens tested with a 1 to 2 by weight mixture of toluene/isobutyl acetate. The fluids shall be applied to a filter paper, which, in turn, is applied as shown in Figure 1. As crazing progresses toward the load, move the filter paper or wrap down the specimen. Exposure time to each fluid shall be 30 minutes (+1, -0 minute). At this time, mark the position of the last craze on the side of the specimen. Measure the distance from the last craze point to the load point using a combination square, modified with a plumb line and leveling bob. Stress to craze value shall be calculated as follows:

$$\text{stress to craze, psi} = \frac{6\ LW}{wt^2}$$

L = length from last craze to load point

W = applied load in pounds

w = specimen width

t = specimen thickness

4.6.13 Accelerated weathering. Specimens from 4.6.9 shall be subjected to accelerated weathering in accordance with Method A of ASTM G 26 for 240 hours. Each test specimen shall be visually examined for conformance to 3.15.2, then subjected to examinations for luminous transmittance and haze, 4.6.9. Specimens shall also be subjected to the warpage examination in 4.6.13.1.

4.6.13.1 Warpage after accelerated weathering. The accelerated weathering specimens shall be conditioned on a plane surface. After conditioning, the specimen shall be measured for warpage by determining the greatest distance from a straight edge connecting diagonally opposite corners to the near surface of the plastic. This distance may be measured by means of dial micrometer, thickness gage or any other device having an accuracy of 0.001 inch (.003 cm). The warpage reported shall be the maximum value, not an average (see 3.13.3).

4.6.14 Thickness. The thickness shall be measured by means of a dial micrometer, thickness gage or any other device having an accuracy of 0.001 inch (.003 cm). Thickness shall conform to 3.3.2.

4.6.15 Long term water absorption. Three 1 by 2 by 0.125 inch (2.5 by 5.0 by 0.3 cm) thick specimens shall be vacuum dried at 70°$\pm$ 1°C (158°$\pm$ 2°F) for 72 $\pm$ 1 hours. The dry weight (W_1) shall be determined to the nearest milligram followed by immediate immersion in water at 60°C $\pm$ 1°C (140°$\pm$ 2°F)

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for a period of 25 days +1, -0 hours. The specimens shall be removed from the water, immediately surface dried using a soft cloth and re-weighed (W_2). Long term absorption shall be determined as follows:

$$\% \text{ Long term absorption} = \frac{W_2 - W_1}{W_1} \times 100$$

5. PACKAGING

5.1 Preservation. Preservation shall consist of a protective covering on both sides of the plastic sheet. The protective covering shall be a suitably adhered paper or film that can be readily removed without injury or damage to the plastic surface (see 3.15). The covering shall adequately protect the surfaces from scratches or damage during shipment.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2.1). Containers shall be suitable for maintaining the plastic in its original condition. All internal loads shall be supported to avoid damage to surfaces by cleats, etc. An instruction sheet shall be included in each individual exterior container (see 3.14). All plastic sheets shall be packed as far as practicable by size and thickness.

5.2.1 Level A. Plastic sheets, preserved as specified in 5.1, shall be packed for shipment in overseas type containers conforming to PPP-B-585 (Class 3), PPP-B-591, PPP-B-601 or PPP-B-621 (Class 2, Style 2). Plywood shall conform to NN-P-530, Type I or II, Grade 4 of Group A. Plywood shall be treated in accordance with TT-W-572. Containers conforming to PPP-B-591 or PPP-B-601 shall be modified to the extent that solid wood ends and sides, in lieu of the cleated type, shall be used. Wirebound boxes conforming to PPP-B-585 shall be provided with fiberboard liners having a minimum Mullen test of not less that 275 pounds. All cleats shall be flush and containers shall be designed in a manner which will insure even weight distribution over the entire bearing surface when the sheets are stored. Containers shall be closed and strapped in accordance with the applicable container specification and appendix thereto. Containers conforming to PPP-B-585, PPP-B-591 or PPP-B-621 shall be provided with a case liner fabricated and sealed in accordance with MIL-L-10547. The gross weight of the shipping container, when packed for shipment, shall not exceed approximately 200 pounds, except when the weight of a single packed sheet exceeds this limitation.

5.2.2 Level B. Plastic sheets, preserved as specified in 5.1, shall be packed for shipment in domestic type containers conforming to PPP-B-585, PPP-B-591, PPP-B-601 or PPP-B-621 (Class 1, Style 2). Containers conforming to PPP-B-591 or PPP-B-601 shall be modified to the extent that solid wood ends and sides in lieu of cleated type shall be used. Wirebound boxes conforming to PPP-B-585 shall be provided with fiberboard liners having a minimum Mullen test of not less than 275 pounds. All cleats shall be flush and containers shall be designed in a manner which will insure even weight distribution over closed and strapped in accordance with the applicable container specification and appendix thereto. The gross weight of the shipping container, when packed for shipment, shall not exceed 500 pounds, except when the weight of a single packed sheet exceeds this limitation.

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5.2.3 Level C. Packages which require over-packing for acceptance by the carrier shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet Uniform Classification Rules or regulations of other common carriers as applicable to the mode of transportation.

5.3 Marking of shipment.

5.3.1 Marking of individual sheets. The protective covering (see 5.1) of each plastic sheet shall be distinctly marked, at intervals of 1 foot (0.3 m), with the following:

* a. Specification number (including Type and Class).

b. Nominal thickness.

c. Manufacturer's designation or code for the product.

In addition, the national stock number (NSN) shall be marked a minimum of one time on each protective covering.

5.3.2 Shipping containers. Each shipping container shall be marked to indicate method of packing and, in addition, shall be marked in accordance with the requirements applicable to the individual services, as specified in MIL-STD-129. The identification shall be composed of the following information listed in the order shown:

* Stock number or other identification number as specified in the purchase order or contract.
PLASTIC SHEET, ACRYLIC MODIFIED
MIL-P-8184E
Type, Class.
Quantity.
Size and thickness.
Color of plastic sheet (when required).
Date of manufacture (month and year).
Stored items shall be issued by their date of manufacture (i.e., older items shall be issued first).
Contract or order no.
Name of manufacturer.
Name of contractor (if different from manufacturer).

NOTE: The contractor shall enter the NSN specified in the purchase document or as furnished by the acquiring activity. When the NSN is not provided or available from the acquiring activity, leave space therefore and enter the number or other identification as provided by the acquiring activity.

5.3.3 Additional marking. In addition to all other required markings, the top of each shipping container shall be marked with the following:

SHIP AND STORE - TOP UP

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6. NOTES

* 6.1 Intended use. This modified acrylic plastic sheet is primarily intended for transparent areas on aircraft where a material with good optical, formability, crazing resistance, outdoor weathering and heat resistance properties is required. Type II materials shall not be used in stretching operations to produce sheets and forms conforming to MIL-P-25690.

6.1.1 Drawing references. Reference of this specification on drawings or in product specifications may be made in order to specify the flat material to be used in the fabrication of a part. It must be clearly understood that the requirements herein apply to the flat material prior to forming or fabrication and not necessarily to the material in its final form. The thickness of the material (see 1.2) must be specified on the drawings or product specification.

* 6.2 Ordering data.

6.2.1 Acquisition documents. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Type and class.

c. Quantity.

d. Dimensions.

e. Color, spectral transmittance and haze, when required.

f. Selection of applicable levels of packing.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List (QPL 8184) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Washington, DC 20361; however, information pertaining to qualification of products and letter of authorization for submittal of sample may be obtained from the Commander, Naval Air Development Center, Warminster, PA 18974, Attn: Aircraft and Crew Systems Technical Directorate (Code 6061).

6.4 Solubility. These acrylic plastic products may be attacked or undergo swelling in ketones, esters, aromatic hydrocarbons and chlorinated hydrocarbons.

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6.5 Additional optical requirements. In the event material conforming to this specification is used for specialized optical purposes, more rigid optical requirements may be specified in the detail product specification. In this event, the requirements of the detailed specification prevail.

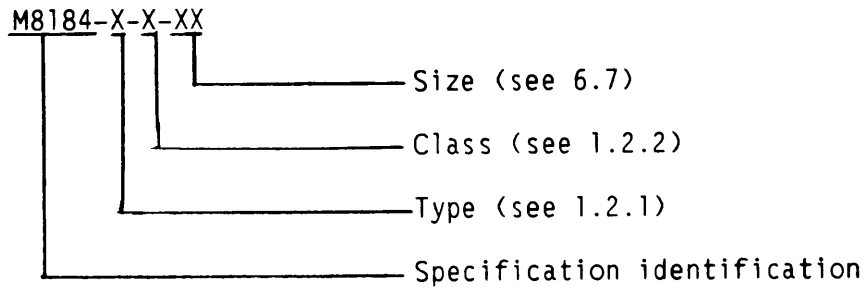
6.6 Variation of physical properties with temperature. Many physical properties of this material vary with temperature. This fact should be considered by designers, engineers, draftsmen and prospective users of the material.

*6.7 Sizes. Sizes previously direct purchased by the Government are:

Size Designation	Width	Length	Thickness
1	18	60	0.060
2	18.25	24	0.250
3	20	30	0.375
4	20.13	48	0.060
5	36	36	0.313
6	36	36	0.750
7	36	48	0.080
8	36	48	0.100
9	36	48	0.125
10	36	48	0.150
11	36	48	0.187
12	36	48	0.220
13	36	48	0.250
14	36	48	0.375
15	36	48	0.500
16	48	48	0.250

The above sizes are those which have been purchased by the Government for direct use and do not include all the available sizes of these materials used by the aircraft transparency industry.

6.8 Part numbering. Part numbering for Government cataloging purposes shall be as follows:



Example: Type I, Class 1, sheet 36 by 48 by 0.080 inches thick shall be cataloged as M8184-1-1-07.

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* 6.9 Keyword listing.

Acrylic sheet
Aircraft transparency
Craze resistance
Heat resistance
Qualification

6.10 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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Navy - AS
Air Force - 20

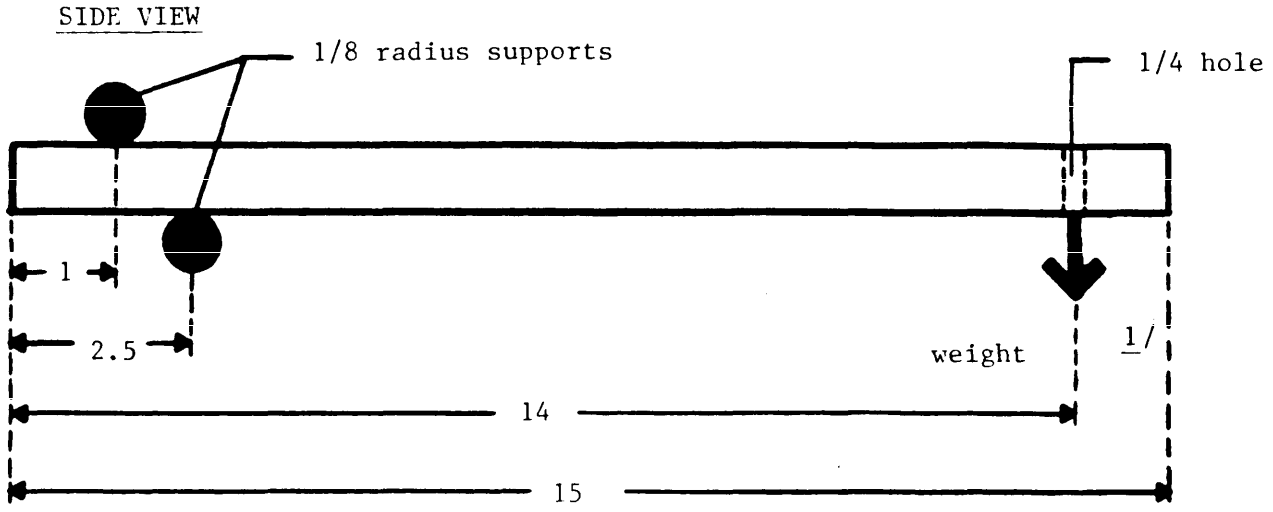
Review activities:
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Navy - EC
DLA - GS

User activities:
Army - ME
Navy - SH, MC

Preparing activity:
Navy - AS

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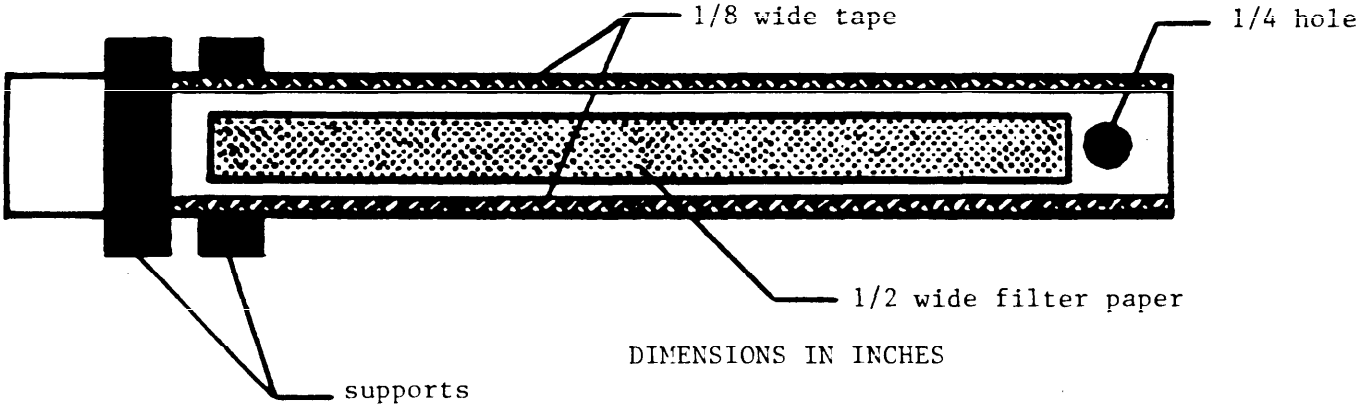
1/ Applied weight as follows:

$$W = \frac{w\ t^2\ \sigma}{6\ L}$$

Where:

W = applied weight (pounds)
w = specimen width (1 inch)
t = specimen thickness (inch)
σ = outer fiber stress (4000 psi)

L = distance from fulcrum to load point (11.5 inch)



* FIGURE 1. Crazing apparatus configuration.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-8184E	PLASTIC SHEET, ACRYLIC, MODIFIED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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